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 Studierichting: Informatica
 Oefeningenreeks 7A nr. 18.2

$$f(x, y, z) = \begin{cases} \frac{xyz}{x^2 + y^2 + z^2} & \text{als } (x, y, z) \neq (0, 0, 0) \\ 0 & \text{als } (x, y, z) = (0, 0, 0) \end{cases}$$

Continuïteit:

Stel $x = r \sin \phi \cos \theta$, $y = r \sin \phi \sin \theta$ en $z = r \cos \phi$

$$\begin{aligned} \lim_{(x,y,z) \rightarrow (0,0,0)} f(x, y, z) &= \lim_{r \rightarrow 0} \frac{(r \sin \phi \cos \theta)(r \sin \phi \sin \theta)(r \cos \phi)}{r^2 \sin^2 \phi \cos^2 \theta + r^2 \sin^2 \phi \sin^2 \theta + r^2 \cos^2 \phi} \\ &= \lim_{r \rightarrow 0} \frac{r^3 \sin^2 \phi \cos \phi \sin \theta \cos \theta}{r^2 (\sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + \cos^2 \phi)} \\ &= \lim_{r \rightarrow 0} \frac{r^3 \sin^2 \phi \cos \phi \sin \theta \cos \theta}{r^2 (\sin^2 \phi + \cos^2 \phi)} \\ &= \lim_{r \rightarrow 0} \frac{r^3 \sin^2 \phi \cos \phi \sin \theta \cos \theta}{r^2} \\ &= \lim_{r \rightarrow 0} r (\sin^2 \phi \cos \phi \sin \theta \cos \theta) = 0 \\ &\Rightarrow f \text{ is continu in } (0, 0, 0) \end{aligned}$$

Partiële afgeleiden:

$$\begin{cases} \frac{\partial f}{\partial x}(0, 0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(\lambda, 0, 0) - f(0, 0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \\ \frac{\partial f}{\partial y}(0, 0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(0, \lambda, 0) - f(0, 0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \\ \frac{\partial f}{\partial z}(0, 0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(0, 0, \lambda) - f(0, 0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \end{cases}$$

$\Rightarrow f$ is partieel afleidbaar in $(0, 0, 0)$

Afleidbaarheid:

Stel $\bar{h} = (h_1, h_2, h_3)$

$$\begin{aligned} Df(\bar{0}, \bar{h}) &= \lim_{\lambda \rightarrow 0} \frac{f(\lambda h_1, \lambda h_2, \lambda h_3) - f(0, 0, 0)}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{(\lambda h_1)(\lambda h_2)(\lambda h_3)}{\lambda^2 h_1^2 + \lambda^2 h_2^2 + \lambda^2 h_3^2} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^3 h_1 h_2 h_3}{\lambda (\lambda^2 h_1^2 + \lambda^2 h_2^2 + \lambda^2 h_3^2)} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^3 h_1 h_2 h_3}{\lambda^3 (h_1^2 + h_2^2 + h_3^2)} = \frac{h_1 h_2 h_3}{h_1^2 + h_2^2 + h_3^2} \end{aligned}$$

$\Rightarrow f$ is afleidbaar in $(0, 0, 0)$ (in alle richtingen $\bar{h}$)

Differentieerbaarheid:

Voor differentieerbaarheid in $(0, 0, 0)$ moet gelden:

$$Df(\bar{0}, \bar{h}) = \begin{pmatrix} \frac{\partial f}{\partial x}(0, 0, 0) & \frac{\partial f}{\partial y}(0, 0, 0) & \frac{\partial f}{\partial z}(0, 0, 0) \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = 0$$

$$Df(\bar{0}, \bar{h}) = \frac{h_1 h_2 h_3}{h_1^2 + h_2^2 + h_3^2} \neq 0$$

$\Rightarrow f$ is niet differentieerbaar in $(0, 0, 0)$

References